

Osijek Competitive Programming Camp 2025 Winter

Day 9: New World

Solutions

Problem & Solution Author: Aeren

February 23, 2025

A Saki and False Minoshiro

For non-negative integer p , let's say p is a **period** of a sequence s if $s_i = s_{i+p}$ for all $0 \leq i < N-p$. We'll define the **autocorrelation** of a string s , denoted by $A(s)$, as the binary string of length N where $A(s)_p = 1$ if and only if p is a period of s .

We introduce two properties related to a binary string being an autocorrelation.

Definition.

A binary string v of length N satisfies the **forward propagation rule (FPR)** if for each $0 \leq p < q < N$ with $v_p = v_q = 1$, $v_{p+i(q-p)} = 1$ for all $0 \leq i$ with $p + i(q-p) < N$.

A binary string v of length N satisfies the **backward propagation rule (BPR)** if for each $0 < p < q \leq 2p < 2N$ with $v_p = 1$ and $v_{2p-q} = 0$, $v_{p-i(q-p)} = 0$ for each $1 \leq i \leq \lfloor (N-p)/(q-p) \rfloor$ with $p - i(q-p) \geq 0$.

Note that if a binary string satisfies FPR or BPR, so will any of its substrings.

Theorem. GCD Rule

Let v be a non-empty binary string of length N satisfying $v_0 = 1$ and both FPR and BPR. Then for all $1 \leq p < q < N$ with $v_p = v_q = 1$ and $p + q \leq N + \gcd(p, q)$, $v_{\gcd(p, q)} = 1$.

Proof.

Since we're only working with indices that is a multiple of $\gcd(p, q)$, we can replace the entire string with only those whose index is a multiple of $\gcd(p, q)$, reducing to the case of $\gcd(p, q) = 1$.

Now, it's enough to prove the theorem for the prefix of length $p + q - 1$. Note that the theorem implies that v is now a constant binary string of 1s by the FPR.

Let's induct on q . The base case of $p = 1, q = 2$ is trivial. Suppose that the theorem has been proven for all $p' < q'$ with $\gcd(p', q') = 1$ and $q' < q$. Let $q = m \cdot p + r$ with $1 \leq r < p$. Since $w = (v_{m \cdot p}, v_{m \cdot p + 1}, \dots, v_{N-1})$ satisfies both FPR and BPR (because it is a suffix of v) and $w_0 = v_{m \cdot p} = 1$ (because if p is a period, all multiples of p is a period as well), if $r \geq 2$, w is a constant of 1s if $r \geq 2$ by the induction hypothesis, and if $r = 1$, w is a constant of 1s by the FPR. We now repeatedly apply BPR to conclude that no position to the left of $v_{m \cdot p}$ contains 0.

□

Let $BP(v)$ be the minimum positive integer p with $v_p = 1$, or N if there is no such p .

Definition. Predicate Ξ

Let v be a binary string of length N , and let $p = BP(v)$. v satisfies the predicate Ξ if either v is empty, or satisfies the following constraints:

1. $v_0 = 1$

2. We consider this case only when $N \geq 2p$.

Let $r = (N \bmod p) + p$. In this case, we must have $v_i = 0$ for all $1 \leq i < n - r$ except at multiples of p , in which case $v_i = 1$. Let w be the suffix of v of length r . Then we must also have all of the followings.

2.a $w_p = 1$ or $r = p$

2.b If $BP(w) < p$, then $BP(w) > (r - p) + \gcd(p, BP(w))$

2.c w satisfies Ξ .

3. We consider this case only when $N < 2p$.

Let $r = n - p$. In this case, we must have $v_i = 0$ for all $1 \leq i < p$. Let w be the suffix of v of length r . Then w must satisfy Ξ .

We now introduce the main theorem we need to solve the problem.

Theorem. Let v be a non-zero binary vector of length N . Then the following four statements are equivalent.

- (a) v is an autocorrelation of a binary string.
- (b) v is an autocorrelation of some string.
- (c) $v_0 = 1$ and v satisfies FPR and BPR.
- (d) v satisfies Ξ .

Proof.

(a) $\rightarrow$ (b)

This case is trivial.

(b) $\rightarrow$ (c)

Assume that $v = A(Z)$ for some string Z . By the definition, $v_0 = 1$.

1. Proof of FPR

Let $p < q$ be indices with $v_p = v_q = 1$. Since p is a period of Z , we can write $Z = XY$ where $|X| = p$ and Y is a prefix of Z . Since q is a period of Z , $q - p$ must be a period of Y . Therefore, Y can be decomposed into form $UU \cdots UV$ where $|U| = q - p$ and V is a proper prefix of U . We now obtain that $p + i(q - p)$ is a period of Z for all $i \geq 0$ with $p + i(q - p) < N$.

2. Proof of BPR

Let $p < q \leq 2p$ be indices with $v_p = v_q = 1$ and $v_{2p-q} = 0$. Since p is a period of Z , we can write $Z = XWY$ where $|X| = 2p - q$, $|W| = q - p$ and Y is a prefix of Z . Since q is a period of Z , $q - p$ must be a period of Y . Therefore, Y can be decomposed into form

$UU \dots UV$ where $|U| = q - p$ and V is a proper prefix of U . The condition $v_{2p-q} = 0$ implies that $W \neq U$.

Now assume that there exists an $2 \leq i \leq \lfloor (n - p)/(q - p) \rfloor$ with $p - i(q - p) \geq 0$ and $v_{p-i(q-p)} = 1$. Then we can write $Z = TYR$ with $|T| = p - i(q - p)$. However, we have $TU \dots UVR = XWU \dots UV$, and one of U must coincide with W , which is a contradiction. Therefore, there is no such i , proving BPR.

(c) $\rightarrow$ (d)

We use induction on N . The base case of $N = 1$ is trivial. Now assume the statement is proved for all $1 \leq N' < N$. Note that all the definitions of parameters below are from the definition of Ξ .

1. Case $N \geq 2p$

From the GCD rule, we know that any t with $v_t = 1$ must either be a multiple of p , or $t > p(\lfloor N/p \rfloor - 1)$. 2.a can be verified with FPR. 2.b can be verified with the fact that all substring of strings with FPR and BPR property has FPR and BPR property, and applying GCD rule after. Finally, 2.c follows from our induction hypothesis.

2. Case $N < 2p$

If $p = N$, $r = 0$, and an empty sequence satisfies Ξ , so we're done. Otherwise, w satisfies FPR and BPR, along with $w_0 = 1$ due to the definition of p , so we apply the induction hypothesis on w and we're done.

(d) $\rightarrow$ (a)

We again use induction on N , but we'll actually prove something stronger: If v is any binary string satisfying Ξ , and W is any binary string whose autocorrelation is w (in the definition of Ξ), whose existence is given by the induction hypothesis, there exists a binary string V with suffix W and autocorrelation v .

1. Case $N \geq 2p$

W has period p , and we can write $W = PR$ where $|P| = p$ and $|W| = r$. We set $V = P \dots PR$ so that $|V| = N$. Clearly, V has all the periods indicated by v , so we just need to show that no additional periods have been introduced. This is equivalent to showing that if $t < n - r$ and $p \nmid t$, t is not a period of V . Suppose there exists such t which is a period of V , and we pick the minimum such t . By the GCD rule, t must be a proper divisor of p . However, such t is always a period of W and $2t \leq p$, so $BP(w)$ divides t , and thus $BP(w)$ divides p , which is a contradiction to the condition $BP(w) < p$.

2. Case $N < 2p$

If W is empty, we have $N = p$, and it's sufficient to set $V = 0 \dots 01$.

Otherwise, our goal is to determine a binary string X such that $V = WXW$ has autocorrelation v . Let $x = |X| = n - 2r > 0$ be its length, and $t = BP(w)$.

(a) Case $x \geq t$

In this case, it's sufficient to set $X = a \dots a$ where a is the complement of the leftmost bit of w . A shift in range $[1, t)$ will cause a mismatch over W , a shift in range $[t, r)$ will make t consecutive characters of W to be placed under X , causing a mismatch, and finally a shift in range $[r, p)$ will cause the first character of W to be placed under X , causing a mismatch. Therefore, the autocorrelation of V is $10 \dots 0w$ as desired.

(b) Case $x < t$

We write $r = m \cdot t + y$ with $0 \leq y < t$.

i. Case $m > 1$

By the GCD rule, any period q of W which is not a multiple of t must satisfy $q > (m-1)t + y$. First, note that a shift of V in range $[r, n-r)$ cannot result in a match no matter what X is, since it would violate the constraints that t is a basic period of w .

Let z be the smallest shift in $[1, r)$ which can be a period of $Z = WXW$. By the GCD rule, $r+x$ is divisible by z which is a period of W . Since $r \leq r+x < r+t$, $r+x = (m+1)t$ must hold. Suppose there is some period q of W dividing $r+x$, which is not a multiple of t . Such q satisfies $(m-1)t < q \leq r$. If $r+x = kq$ ($k \geq 2$), we obtain

$$k(m-1)t + ky < (m+1)t + y$$

and this can only hold when $m = k = 2$. Thus, we have three cases.

A. $r+x$ is not a multiple of t .

In this case, we can arbitrarily set X .

B. $r+x$ is a multiple of t and its multiples, and nothing else.

In this case, we let $X = ? \dots ? a$ where a is the complement of W_{t-1} and $?$ is an arbitrary character. Any shift in range $[1, r)$ which can match must be a multiple of t , and thus it will fail to match over X .

C. $r+x$ is a multiple of t and another period q which is not a multiple of t , in which case we have that $m = 2$ and $r+x = 2q$.

In this case, we must have $3t = 2q$, implying that $t \geq 2$. Now $x = t - y$, and if $y = 0$, then $x = t \geq 2$. Otherwise, from $q = 3y/2 > t + y$, we obtain $t > 2y$, and $x = t - y > 2y - y = y \geq 1$, and thus $x \geq 2$ again.

Therefore, we conclude that $|X| \geq 2$. Now we set X_{x-1} to the complement of W_{t-1} , and X_{x-2} to the complement of W_{q-2} , and rest to arbitrary. Such X will cause any shift in $[1, r)$ to mismatch.

ii. Case $m = 1$

We have $r = t + y$ and any other period q of W must satisfy $q > t$. A multiple $ku = r + x$ of some period u of W can be in the interval $[t + y, 2t + y)$ only if $kt < 2t + y$, and so we must have $k = 2$. Therefore, at most one such u exists.

If such u exists, we set the last character of X to the complement of W_{u-1} . Otherwise, X can be arbitrary.

□

Since $M \geq 2$, the set of autocorrelation of interest is independent of M . Furthermore, the definition of Ξ gives an efficient generation algorithm of all autocorrelations. It turns out that there are only 35200 of them for $N = 100$. Generate all of them, and we need to do a dynamic programming over them to obtain the answer.

For an autocorrelation v , the **degree of freedom** of v , denoted by $F(v)$, is the number of positions we can freely set in the original string. More specially, each period in v defines a set of equality among characters of the original string, and $F(v)$ is the number of final equivalence classes. We can assign any of the M character to each free characters, but this will contains strings with autocorrelation strictly containing v . Thus, we need to process autocorrelations in decreasing order of 1s, and set

$$DP[v] = M^{F(v)} - \sum_{u \text{ is a superset of } v} DP[u]$$

where $DP[v]$ is the number of strings with autocorrelation equal to v . It is now trivial to compute the answer from DP .

Time Complexity: $O(\kappa(N)^2)$ where $\kappa(N)$ is the number of autocorrelations of length N

B Saki and Sewage System

Theorem.

Such assignment of capacities exist if and only if there exists a path from 0 to $N - 1$ in B such that the corresponding set of edges does not connect 0 and $N - 1$ in A .

Proof.

If such path exists, we set the capacity of all edges in the path to 1 and the rest to 0, and we're done.

If such path does NOT exist, consider a maximum flow in B . We know that the flow can be decomposed into F unit flows, and some cycles, where F is the amount of flow. For each unit flow, since it represent a path, the set of edges in that path must connect 0 and $N - 1$ in A . Thus, each unit flow generates at least 1 amount of flow in A , so the maximum amount of flow in A must be equal or greater than F .

□

The strategy now is clear. We try all possible way to separate $0, \dots, N - 1$ into two sets $0 \in S$ and $N - 1 \in T$, add all edges in A such that it connects the vertices in the same set in A , and check if it connects 0 and $N - 1$ in B .

Time Complexity: $O(2^N \cdot M)$

C Saki and Separation

Let

$$F(X, Y) = \sum_{i=0}^{N-1} \sum_{j=0}^{M-1} a_{i,j} X^i Y^j$$

$$G(X, Y) = \sum_{i=0}^{N-1} \sum_{j=0}^{M-1} b_{N-1-i, M-1-j} X^i Y^j$$

where $a_{i,j} = 1$ if $S_{i,j} = A$, otherwise $a_{i,j} = 0$, and $b_{i,j} = 1$ if $S_{i,j} = B$, otherwise $b_{i,j} = 0$.

$$H(X, Y) = F(X, Y) \cdot G(X, Y)$$

$$= \sum_{i=0}^{N-1} \sum_{j=0}^{M-1} \left(\sum_{k+l=i} \sum_{m+n=j} a_{k,m} b_{N-1-l, M-1-n} \right) X^i Y^j$$

$$= \sum_{i=0}^{N-1} \sum_{j=0}^{M-1} \left(\sum_{k-l=i-(N-1)} \sum_{m-n=j-(M-1)} a_{k,m} b_{l,n} \right) X^i Y^j$$

Note that $\sum_{k-l=i-(N-1)} \sum_{m-n=j-(M-1)} a_{k,m} b_{l,n}$ is 0 or not depending on whether there exists a pair of students from A and B at the same cell if we change the position of each students in A by $(N - 1 - i, M - 1 - i)$. With this information, we can dfs starting from the displacement of $(0, 0)$. Thus, we just have to find the product of F and G , which can be done with 2D FFT.

Time Complexity: $O(N \cdot M \cdot \log(N + M))$

D Saki and Railway Construction

Using Kirchhoff's theorem,

$$T_N \stackrel{\text{def}}{=} f(X) = \det \begin{bmatrix} X & -1 & 0 & 0 & \cdots & 0 & 0 \\ -1 & 2X & -1 & 0 & \cdots & 0 & 0 \\ 0 & -1 & 2X & -1 & \cdots & 0 & 0 \\ 0 & 0 & -1 & 2X & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 2X & -1 \\ 0 & 0 & 0 & 0 & \cdots & -1 & 2X \end{bmatrix}$$

Here, $T_N \in \mathbb{Z}[X]$, or we may replace $\mathbb{Z}[X]$ with any of the ring containing $\mathbb{Z}[X]$. We'll also denote the $GF(M)$ -function version (the evaluation mapping over $GF(M)$) as $\overline{T_N} : GF(M) \rightarrow GF(M)$, or we may replace $GF(M)$ with any field containing $GF(M)$.

From the definition,

- $T_2 = 2X^2 - 1$
- $T_3 = 4X^3 - 4X$
- For $N \geq 4$, $T_N = 2X \cdot T_{N-1} - T_{N-2}$

The recurrence relation can be derived by iterating over which element in the last row to use in the determinant computation.

By applying recurrence relation in reversed order, we also define $T_1 = X$, $T_0 = 1$, $T_{-N} = T_N$.

Note the similarity between the recurrence formula and the cosine multi-angle formula: for arbitray real number θ ,

$$\cos(N\theta) = 2 \cos(\theta) \cos((N-1)\theta) - \cos((N-2)\theta)$$

With induction, one can easily derive the following identity.

$$T_N(\cos(\theta)) = \cos(N\theta)$$

(Actually, this formula is the usual definition of T_N , the N -th Chebyshev polynomial)

Note that the above identity is over $\mathbb{R}$, which contains $\mathbb{Z}$. We switch to $\mathbb{C}$, then apply $\cos(\theta) = \frac{e^{\pi i} + e^{-\pi i}}{2}$ to obtain

$$T_N\left(\frac{e^{\pi i} + e^{-\pi i}}{2}\right) = \frac{e^{\pi i N} + e^{-\pi i N}}{2}$$

Therefore, the following identity in $\mathbb{C}(X)$ holds since it is true for uncountably many substitutions of X .

$$T_N\left(\frac{X + X^{-1}}{2}\right) = \frac{X^N + X^{-N}}{2}$$

Another identity we need is the following.

$$T_a \circ T_b = T_{ab}$$

This can be seen by $T_a(T_b(\cos(\theta))) = T_a(\cos(b\theta)) = \cos(ab\theta) = T_{ab}(\cos(\theta))$.

These are the main properties we need to solve this problem.

The problem asks for minimum positive integer K where $\overline{T_N}^{\circ K} = \overline{T_N^K} = \overline{T_1}$.

Case $M = 2$

we have the recurrence $T_a = T_{a-2}$, so for even N , answer is -1 , and for odd N , answer is 1 .

Case $M \geq 3$

We need to have $\overline{T_N^K}(x) = x$ for all $x \in GF(M)$, but this is little hard to work with, as we haven't found a concise expression for $T_N(X)$. Instead, we'll work with $T_N\left(\frac{X+X^{-1}}{2}\right) = \frac{X^N+X^{-N}}{2}$.

Theorem.

1. For all $x \in GF(M)$, there exists at least one $t \in GF(M^2)^\times$ with $x = \frac{t+t^{-1}}{2}$, and the order of all such t divides $M+1$ or $M-1$.
2. For arbitrary $t \in GF(M^2)^\times$ with order of t dividing $M+1$ or $M-1$, $\frac{t+t^{-1}}{2} \in GF(M)$.

Proof.

1. By the Fermat's little theorem, $x^M = x \leftrightarrow t^M + t^{-M} = t + t^{-1}$, and $t = (t^M \text{ or } t^{-M}) \leftrightarrow t^{M-1} = 1 \text{ or } t^{M+1} = 1$.
2. $(t + t^{-1})^M = t^M + t^{-M} = \begin{cases} t + t^{-1} & \text{if the order of } t \text{ divides } M-1 \\ t^{-1} + t & \text{if the order of } t \text{ divides } M+1 \end{cases}$

Since $t + t^{-1}$ is fixed by the Frobenius endomorphism, $t + t^{-1} \in GF(M)$.

□

When does $a, b \in \mathbb{Z}$ satisfies $\overline{T_a} = \overline{T_b}$?

We need to check $\overline{T_a}(x) = \overline{T_b}(x)$ for all $x \in GF(M)$, and the above theorem implies that it's equivalent to checking $\overline{T_a}\left(\frac{t+t^{-1}}{2}\right) = \frac{t^a+t^{-a}}{2} = \overline{T_b}\left(\frac{t+t^{-1}}{2}\right) = \frac{t^b+t^{-b}}{2}$ for all $t \in GF(M^2)^\times$ with order dividing $M+1$ or $M-1$.

Note that $M-1$ can be achieved by picking a generator of $GF(M)^\times$ and $M+1$ can be achieved by r^{M-1} where r is a generator of $GF(M^2)^\times$.

Therefore

$$\begin{aligned} \frac{t^a + t^{-a}}{2} &= \frac{t^b + t^{-b}}{2} && \text{for all such } t \\ \Leftrightarrow t^a = t^b \vee t^a = t^{-b} && \text{for all such } t \\ \Leftrightarrow t^{a+b} = 1 \vee t^{a-b} = 1 && \text{for all such } t \\ \Leftrightarrow a = \pm b \pmod{M+1} \wedge a = \pm b \pmod{M-1} \\ \Leftrightarrow a = b, -b, Mb, -Mb \pmod{\frac{M^2-1}{2}} \end{aligned}$$

So $\overline{T_{NK}} = \overline{T_1}$ if and only if $N^K = 1, -1, M, -M \pmod{\frac{M^2-1}{2}}$. For each of these 4 cases, it can be solved in $O(\sqrt{M})$ times with discrete log and CRT.

Time Complexity: $O(\sqrt{M})$

Alternatively, find the order p of N modulo $\frac{M^2-1}{2}$, and the answer is either $\frac{p}{2}$ (only valid for even p), or p . So check if $N^{\frac{p}{2}} = 1, -1, M, -M \pmod{\frac{M^2-1}{2}}$.

Time Complexity: $O\left(\text{Factorize}\left(\frac{M^2-1}{2}\right)\right)$

E Saki and Hope

First, construct a 220 by 220 grid by adding all points $(x, y, 1)$ with $0 \leq x, y < 220$ to the answer.

Then consider a set of distinct lines of size $100000 - 220^2$ such that it passes as much of the points in the grid as possible. For each line L in the set, add the point represented by the normal vector of the plane containing $(0, 0, 0)$ and the line. This way, each point generated by a line will have large number of perpendicular points with the points on the grid. This will generate $O(N^{4/3})$ number of points. The model solution generates 1418266 pairs.

Note: it seems lots of teams found construction better than the model by multiple fold.

F Saki and Summer Festival

We iterate over all pairs $0 \leq i < j < N$ of points, and count the number of set having i, j as its smallest and second smallest index.

For fixed i and j , we iterate over all $k > j$, and compute the radius of the circle passing through i, j, k , unless they're colinear. We can group k -s which, together with i and j , generates the same circle by checking that

1. corresponding radius is equal, and
2. (if the angle ikj is not $\frac{\pi}{2}$) the value $[\angle ikj \text{ is acute}] \oplus [ikj \text{ is counterclockwise}]$ is equal.
Here, for a proposition P , $[P] = 1$ if P is true, and 0 otherwise.

Note that given three points p, q, r which are not colinear, the radius of the circle passing through all of them is

$$\frac{|p - q| \cdot |q - r| \cdot |r - p|}{|(q - p) \times (r - p)|}$$

We can do all the computation with integer by storing the irreducible fraction of the square of radius.

After grouping with the above criteria, for each group of size x , we add $2^x - 1$ to the answer.

Time Complexity: $O(N^3 \log(N))$

G Saki and Cantus

We need to find the sum of Euler totient function for non-zero Gaussian integers with norm equal or less than N . I will not go into detail about Gaussian integers, as there are lots of freely available explanations, such as Wikipedia.

Euler totient function for Gaussian integer x , with prime factorization $x = u \cdot p_0^{e_0} \cdots p_{k-1}^{e_{k-1}}$, is given by the following formula.

$$\phi(x) = \prod_{i=0}^{k-1} (\text{norm}(p_i) - 1) \text{norm}(p_i)^{e_i-1}$$

Note that $\{1, i, -1, -i\}$ is the set of units of Gaussian integers, and they represent the rotation in the complex plane by $0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}$. We'll compute the desired sum only for Gaussian integers x with $\text{real}(x) > 0$ and $\text{imag}(x) \geq 0$, and multiply the answer by 4 at the end.

The model solution is just the standard Min25 sieve where you replace the sequence of primes in $\mathbb{Z}$ with the primes of Gaussian integers, and instead of computing internals with value up to distinct values of $\lfloor N/K \rfloor$, it computes internals with norm up to distinct values of $\lfloor N/K \rfloor$. Now, the algorithm ~~magically~~ mathematically works for Gaussian integers.

Time Complexity: $O(N^{3/4})$

Note that $O(N^{2/3})$ is possible, but it's not needed to solve this problem.

H Saki and Tree

We first filter out every number that is not a prime or a Carmichael number by running Fermat test with 2 bases. Next, we filter out primes by noticing that all 5 numbers are a multiple of either 7 or 17. The formula model solution implements is

$$(2^{X-1} \bmod X = 1) \wedge (23^{X-1} \bmod X = 1) \wedge (X \bmod 7)(X \bmod 17) = 0$$

Note that $K = 1$ can be implemented with $0^{(K-1)^2}$. The model solution generates a tree with 123 nodes.

I Saki and Encoding

One of the remaining node must be $N - 1$.

Consider the path from $N - 1$ to $N - 2$. The other remaining node must be the node following $N - 1$ in the path. This can be found in $O(1)$

Time Complexity: $O(N)$

J Saki and Decryption

First, we can exclude cases like $p = q$ or $2p - 1 = q$, as these are easy to check with binary search.

We'll compare two domains in parallel: $\mathbb{Z}/N\mathbb{Z}$ and $GF(2p - 1)$. $\mathbb{Z}/N\mathbb{Z}$ is where the actual computation will take place, and $GF(2p - 1)$ will be the domain where the computation is meaningful.

1. Generate a random monic polynomial f of degree 4. Since the number of monic irreducible polynomial over $GF(2p - 1)$ of degree 4 is

$$\frac{1}{4} \left((2p - 1)^4 - (2p - 1)^2 + 1 \right)$$

, the probability that f is irreducible over $GF(2p-1)$ is about $\frac{1}{4}$. In such case, $GF(2p-1)[X]/\langle f \rangle$ is isomorphic to $GF((2p-1)^4)$.

2. Generate a random non-zero element $r \in (\mathbb{Z}/N\mathbb{Z})[X]/\langle f \rangle$. If f is irreducible over $GF(2p-1)$, the probability that r is a generator of $GF((2p-1)^4)$ is

$$\frac{\phi((2p-1)^4 - 1)}{(2p-1)^4 - 1}$$

Since

$$\liminf_{x \rightarrow \infty} \frac{\phi(x)}{x} \ln \ln x = e^{-\gamma}$$

we can estimate the lower bound on the probability as

$$\frac{\phi((2p-1)^4 - 1)}{(2p-1)^4 - 1} \gtrsim \frac{1}{11}$$

3. Compute r^{4N} . Note that $4N = (2p-1)^{\frac{(2p-1)^4-1}{(2p-1)-1}} q^2$, so when f is irreducible and r is a generator of $GF((2p-1)^4)$, r^{4N} has order $(2p-1) - 1$ with probability very close to 1. Therefore, $r^{4N} \in GF(2p-1)^\times$.

This implies that if the result of computation of r^{4N} over $(\mathbb{Z}/N\mathbb{Z})[X]/\langle f \rangle$ is $a + bx + cx^2 + dx^3$, b , c , and d are a random multiple of $2p-1$.

We can compute $2p-1$ by computing $\gcd(N, b)$, and q with binary search. The worst case success probability is bounded from below by about $\frac{1}{44}$, so we expect to run this process 44 times on average at most in the worst case.

Time Complexity: $O(\text{BigintMul}(N) \log(N))$